United International University

B.Sc. in Computer Science and Engineering (CSE)

CSE 4889: Machine Learning

Final Exam: Spring 2026   Time: 2 Hours   Marks: 40

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Answer all of the following questions **sequentially**.

1. a) In the context of back propagation, the error signal for an output neuron is defined as: [5]

$$\delta_o = f' (d - f_o) \Rightarrow \delta_o = f_o (1 - f_o)(d - f_o)$$

Where, d = desired output, and f_o= predicted output. Suppose the weights are initialized such that the net input to the output neuron is very high, resulting in an output $f_o \approx 1$.

(i) If the target value d is 0, explain what happens to the mathematical value of the error signal δ_o as f_o approaches 1.

(ii) Does a high discrepancy between the target (d = 0) and the prediction ($f_o \approx 1$) lead to a large weight update in this specific case? Why or why not?

b) Discuss the role of the smoothing parameter (σ) in a Generalized Regression Neural Network (GRNN). Explain how different values of σ (1, 0.2, and 0.6) affect the model's prediction accuracy, smoothness, generalization ability, sensitivity to noise, and overall performance. [5]

2. Draw the hidden state (s) structure from the following transition probabilities: [10]

$$T_{ij} = \begin{pmatrix} 0.3 & 0 & 0.7 \\ 0 & 0.7 & 0.3 \\ 0.9 & 0 & 0.1 \end{pmatrix}$$

Observation number of phoneme in the samples for phoneme /eə/ :

$$K_{ij} = \begin{pmatrix} 0.3 & 0.4 & 0.3 \\ 0.8 & 0.1 & 0.1 \\ 0.5 & 0.5 & 0 \end{pmatrix}$$

Observation number of phoneme in the samples for phoneme /æ/:

$$Q_{ij} = \begin{pmatrix} 0.8 & 0.1 & 0.1 \\ 0 & 0.3 & 0.7 \\ 0.4 & 0.6 & 0 \end{pmatrix}$$

Assume that, after phoneme segmentation, the numbers of samples observed in four vertical lines are {4, 2, 3, and 5}. Which phoneme is more likely to have generated this observation sequence: /eə/ or /æ/?

3. Why is a Support Vector Machine (SVM) preferred over a Neural Network? Let us consider the training data as follows: [10]

Input	Output
$I_1=1$	$y_1=1$
$I_2=5$	$y_2=-1$
$I_3=6$	$y_3=1$
$I_4=2$	$y_4=1$
$I_5=4$	$y_5=-1$

Set the kernel parameter, d=2, where the polynomial kernel function with degree d is defined as $k(I, y)=(Iy+1)^d$. Subject to the constraint $C \geq \alpha_i > 0$, where C is set to 7. Classify unseen data $I_6=7$ using SVM where obtained α_i values are as follows:

$$\alpha_1=0,\ \alpha_2=6.5,\ \alpha_3=4.5,\ \alpha_4=3.5,\ \alpha_5=8.9$$

4. a) Analyze the stochastic neuron model for pseudo temperatures T=0, 500, and 90,000 °F, using the following single-layer neural network parameters: [5]

Input	Weight
$I_1=1$	$W_1=0.5$
$I_2=0$	$W_2=0.3$
$I_3=2$	$W_2=-0.3$

b) Why does a Recurrent Neural Network (RNN) perform better than a Multi-Layer Perceptron? Consider the following training dataset: [5]

Input (I_1)	Input (I_2)	Desired Output (d)
5	4	0
3	3	1
6	2	0
Input (H)	Input (O)	------
0.1	0.2	

The initial weights and biases for the hidden and output layers are given as follows:

For the hidden layer	For the output layer
$W_{OH}=0.23$	$W_{HO}=0.32$
$W_{HH}=0.11$	$W_{OO}=0.21$
$W_{1H}=-0.23$	$W_{1O}=-0.32$
$W_{2H}=0.21$	$W_{2O}=0.31$
Bias $b_H=0.53$	Bias $b_O=0.42$

Assume:

- Learning rate $\eta=0.005$
- Momentum coefficient $\alpha=0.55$

Draw a schematic diagram of the RNN showing all connections, including input-to-hidden, hidden-to-output, and recurrent links in the hidden and output layers.